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Optimizing Decoy-State Protocols for Practical Quantum Key Distribution Systems

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The decoy-state protocol enables quantum key distribution (QKD) systems to achieve high performance without using the single-photon source. In the dozen years since the decoy-state protocol has been proposed, a series of advances in scheme design, finite-size analysis, system modelling, and parameter estimation has developed. Unfortunately, most advances are based on different starting points and lack a synthesis to figure out the optimal protocol of decoy-state method. Here, the advances in decoy-state method are reviewed and they are synthesized to compare the key-rate performance of 38 decoy-state protocols using the particle swarm optimization (PSO) algorithm. To form a guideline for practical QKD systems, the optimal decoy-state protocols are found with the change of five system parameters (the transmission loss, the block size of postprocessing, the misalignment-error probability, the dark-count probability, and the afterpulse rate). It is expected that this work can answer the questions of what the actual performance is for different decoy-state protocol in various scenarios and which decoy-state protocol is the optimal choice for various QKD systems.

1. Introduction

How to secure the communication between two authorized distant partners, Alice and Bob, against eavesdropper, Eve, is the core problem that the cryptography has tried to answer. The most frequently used method is Alice scrambles the message (plaintext) into an incomprehensible one (ciphertext) for eavesdroppers with a key and algorithm and sends it to Bob via an open channel, then Bob restores the ciphertext to plaintext by a key and algorithm, which is called encryption and decryption. The security of most encryption algorithms is based on the computation complexity which can be threatened by advanced computers, such as Colossus[1] and quantum algorithms.[2] The one-time pad (OTP)[3] is a distinctive encryption algorithm whose security does not depend on computation complexity and has been proved in the last century.[4,5] However, the advantage is not without costs, OTP requires Alice and Bob to share a random single-use key whose length is identical with the size of ciphertext, which indicates a problem of secure key distribution and further limits its application.

Quantum key distribution (QKD)[6,7] can unconditionally securely share identical random-number strings between the partners via an untrusted channel for OTP, whose security is guaranteed by the principle of quantum physics. It is nearly 40 years since the first QKD protocol, BB84,[6] has been proposed. QKD has matured from a demonstration system with a short communication distance and side-channel loopholes[6] into a high-performance system[8–11] that can satisfy some practical requirements.[12–20]

In the ideal QKD design, the classical information of the random number is loaded into the different quantum states of the single photon. The quantum states can be described by the physical dimensions of a photon such as polarization, phase, and time. Take the quantum states in 2D Hilbert space, for example. There are three sets of states corresponding to three mutually unbiased bases. Each basis contains two orthogonal states which represent the bit 0 and bit 1 of random numbers, and the states of different bases are non-orthogonal. When executing the BB84 protocol, as shown in **Figure 1**, the photon state is prepared by Alice, transmitted in an untrusted channel, and measured by Bob. After the measurement, Bob obtains his key from the measurement outcomes. Then they disclose their basis choices and discard the key generated by mismatched bases. When Eve tries to steal information from photon states, based on the quantum non-cloning theorem,[21] she has to operate the original photon state. However, because the basis of preparation is selected randomly and unknown to Eve, the existence of non-orthogonal states leads to an inevitability of the state change introduced by eavesdropping.

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DOI: 10.1002/qute.202000131

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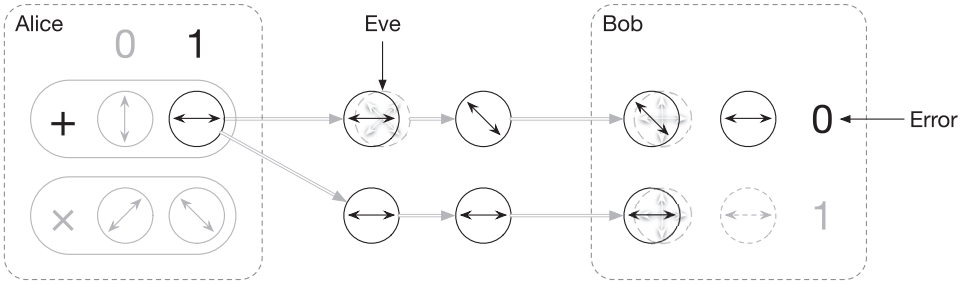


Figure 1. Schematic diagram of BB84 protocol using a single-photon source. Alice prepares a single-photon horizontal polarization state selected from the + basis. When no attacks in the channel, Bob can obtain an identical key using the same basis as Alice. The interception of Eve can change the photon state because she does not know the basis Alice selected leading to an error key observed by Bob.

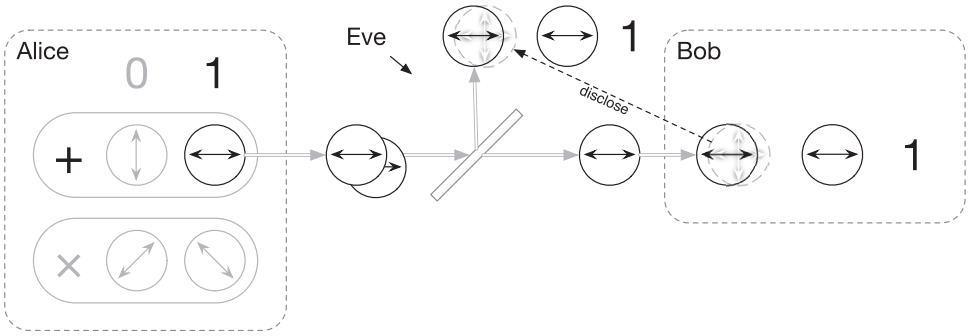


Figure 2. Schematic diagram of PNS attack. Splitting the multi-photon pulse, Eve can withhold one of the photons and send the rest to Bob. Then, she can obtain the same result as Bob without introducing any error, so long as she measures her photon with the same basis as Alice and Bob after they disclose their basis choice.

Therefore, the eavesdropping can be detected according to the statistical properties of measurement results, such as the error rate, which protects the key information from disclosure.

The security of the quantum key is guaranteed by the single photon[22]; however, the single-photon encoding is difficult to realize because the single-photon source is unpractical. The closest leaser source to the practicality is the phase-randomized weak coherent pulses (WCPs)[23] created by the attenuated laser, which inevitably contains multi-photon pulses. The multi-photon pulse is equivalent to the clone of the single photon. Therefore, as shown in **Figure 2**, Eve can withhold part of the pulse using beam splitters and send the rest to Bob. Then, she can measure the pulse after the basis choices are disclosed. Since the photon states of Eve and Bob are identical, she can obtain the same key as Bob without a trace. This attack strategy, called photon-number splitting (PNS) attack,[24,25] poses a serious threat to the security of QKD using practical laser sources.

Fortunately, the single-photon pulses of WCPs are still secure. The key may continue to be generated, as long as Alice and Bob can characterize the yield and error yield of single-photon pulses passing through the channel. Therefore, how to estimate the parameters of single-photon pulses using weak coherent sources determines the tightness of the lower bound of secure key rate.[22]

It is essentially a mathematical problem of solving infinite unknowns with an equation set. Before the decoy-state method,[26–28] the most pessimistic method is attributing all error to the single-photon pulses and assuming all multi-photon pulses can be detected, which leads to a quadric wastage of secure key over the channel loss. The decoy-state method can relatively accurately depict the effect of channel on the single photons by sending pulses with different photon number distributions. Specifically, the number of the unknown parameters is infinite because there are infinite photon-number states using the weak coherent source. Therefore, an equation set that contains infinite equations can help to get the exact solution, each equation is built by preparing a decoy state, which is the infinite decoy-state method. However, it is impossible to prepare infinite decoy states in real life. Therefore, the finite decoy states degrade the mathematical problem into the indeterminate equations. Fortunately, when the number and photon-number distribution of decoy states are appropriate, we can still find a relatively accurate solution to generate a highly secure key rate. The secure key rate is therefore improved and can meet practical requirements.

The decoy-state method plays a key role in the leaping development of QKD, which frees the QKD from the single-photon source and slows the rate of secure key decay with the channel loss to linearity. For decades, various decoy-state protocols are proposed and demonstrated with respect to security and performance,[29–47] for example, the remarkable record transmission[48] is achieved with the 1-decoy protocol.[49,50] The analyses of statistical fluctuation and finite-size effect have been advanced. Meanwhile, the estimation of the parameters of single-photon pulses has become more accurate, and the optimization of decoy-state protocols has been innovated.

Regrettably, these advances do not belong to the same cladistic branch; an aggregated result is still missing. Moreover, for the

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actual QKD systems with different configurations and environmental parameters, it is not conclusive which scheme is optimal. In this paper, we synthesize the advances in the decoy method, the finite-size analysis based on sampling without replacement,[41] the models of afterpulse and misalignment error,[51,52] and the parameter estimation using the data of mismatch basis,[37] and compare and contrast the secure key rate of the various decoy-state protocols under different system conditions. We hope to provide a reference for a real QKD system to choose the optimal decoy scheme.

2. Protocol

In this section, we provide a detailed description of the protocol and the method of key rate generation we employed. Here, we consider the BB84 protocol, where the quantum states are encoded with bases X and Z. For the decoy encoding, Alice can prepare pulses with different intensities in different bases. For simplification, we only consider the situation that the number of intensity in each basis is between two and three. Therefore, the total number of intensity is between two and six, which we believe is enough to cover the optimal scheme. The intensities of bases X and Z are denoted by $\mu_{X1}, \mu_{X2}, \mu_{X3}$ and $\mu_{Z1}, \mu_{Z2}, \mu_{Z3}$, respectively, and satisfy $0 < \mu_{X1} < \mu_{X2} < \mu_{X3}$ and $0 < \mu_{Z1} < \mu_{Z2} < \mu_{Z3}$. The situation that the intensity can be equal to zero is excluded because it is difficult to see a decoy-state modulator with infinite extinction ratio in high-speed systems.

Here, we use N_{IZ} and *Xlock* to describe the various decoy-state protocols, where N_{IZ} is the number of intensity in Z basis, and *Xlock* is an array and used to describe the intensity scheme in X basis. Specifically, the length of *Xlock* represents the number of intensity in X basis. Moreover, each element of *Xlock* is a natural number between 0 and N_{IZ} corresponding to the relationship between the intensities in X basis and Z basis. The value of the ith element is the sequence number of the decoy states in Z basis. When the value of the element is 0, the corresponding intensity, μ_{Xi}, is independent to the Z basis. When the value of the element is $k (0 < k \leq N_{IZ})$, μ_{Xi} is locked in the intensity of kth decoy state in Z basis, μ_{Zk}. For example, the configuration of $N_{IZ} = 2$ and $Xlock = \{0, 0, 2\}$ represents the decoy scheme that Alice prepares μ_{Z1}, μ_{Z2}, μ_{X1}, μ_{X2}, μ_{X3} and $\mu_{X3} \equiv \mu_{Z2}$.

2.1. Preparation

Alice configures a decoy scheme and generates random numbers to determine her initial key and decoy state. She selects a binary key choice, γ_i, with the uniform probability distribution, and an intensity-basis choice in $\{\mu_{Z1}, \ldots, \mu_{Zn}, \mu_{X1}, \ldots, \mu_{Xm}\}$ with probabilities $p_{\mu_{Z1}}, \ldots, p_{\mu_{Zn}}, p_{\mu_{X1}}, \ldots, p_{\mu_{Xm}}$, where $\sum_{i=1}^{n} p_{\mu_{Zi}} + \sum_{j=1}^{m} p_{\mu_{Xj}} = 1$. Alice records the basis choice a_i, the intensity choice k_i, and the key choice γ_i. Finally, she prepares a quantum state according to the random numbers and sends it to Bob via the quantum channel.

2.2. Measurement

Bob chooses a basis $b_i \in \{X, Z\}$ with probabilities p_x and p_z, where $p_x + p_z = 1$, and performs a measurement in it. Here, we analyze the most commonly used dual-detector scheme, where one of the detectors' responds corresponds an outcome of key 0 or 1, the double-click event corresponds a random key and the no-click event indicates an invalid outcome which should be discarded in the later process. For the first three outcomes, Bob records the key to γ_i', and for the last outcome, he records a ø to γ_i'.

2.3. Basis Sifting

Alice and Bob announce their basis and intensity choices over an authenticated public channel and identify the following sets: $\mathcal{X}_k := \{i : a_i = b_i = X \wedge k_i = k \wedge \gamma_i' \neq \emptyset\}$ and $\mathcal{Z}_k := \{i : a_i = b_i = Z \wedge k_i = k \wedge \gamma_i' \neq \emptyset\}$ for all k, which means Alice and Bob retain the valid outcomes they obtained in the same basis. Then, they check for $\sum_k |\mathcal{Z}_k| := \sum_k n_k^Z \geq n_Z$. They repeat steps 1–3 until the condition is satisfied, at this point $\sum_k |\mathcal{X}_k| := \sum_k n_k^X = n_X$. We denote N as the number of laser pulses sent by Alice until the condition is fulfilled.

2.4. Postprocessing

In the protocol, both bases X and Z can generate secure key, here we take bases Z as an example. The raw key pair in basis Z, $(\mathbf{Z}_A, \mathbf{Z}_B)$, is generated by choosing a random sample of size n_Z, where n_Z is the postprocessing block. Then, Alice and Bob announce the sets $\mathcal{Z}_k$ and compute the corresponding numbers of bit errors, m_k^Z. According to the decoy method and the observers of n_k^Z and m_k^Z, they can estimate the lower bounds of the number of vacuum events s_0^{Z-} and the number of single-photon events s_1^{Z-} and the upper bounds of the single-photon phase error rate $e_{1,p}^{Z+}$. If the $e_{1,p}^{Z+}$ is not less than the predetermined threshold $e_{1,p}^{tol}$, the protocol is aborted, otherwise, they perform the error correction and the privacy amplification. For the error correction, up to λ_{EC} bits of information are revealed. Moreover, $\log_2 1/\varepsilon_{hash}$ of bits are published to perform an error verification for the identical kay pair, where ε_{hash} is the probability that a pair of nonidentical keys passes the error-verification step. Finally, they perform privacy amplification on their keys to extract a secret key pair $(\mathbf{S}_{Z,A}, \mathbf{S}_{Z,B})$, where $|\mathbf{S}_{Z,A}| = |\mathbf{S}_{Z,B}| = l_Z$) is the length of the secure key obtained in basis Z. Similarly, they can also generate a secret key pair $(\mathbf{S}_{X,A}, \mathbf{S}_{X,B})$ with a length of l_X.

3. Secure Key Rate

3.1. Workflow

The simulation of the secure key rate can be divided into three steps. First, simulating the counts and error counts based on a set of system parameters. Second, estimating the counts and error counts of vacuum and single-photon states and calculating the secure key rate using the decoy method and finite-size analysis. Third, the optimization algorithm finds the optimal parameters according to the feedback of the secure key rate. The following sections introduce these steps, respectively.

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3.2. Model

3.2.1. Tendency

Modeling the QKD systems can help with security analysis and high-credibility simulation. Some may need an abstract model for the analysis at the protocol level, some need details to study the imperfections of devices. Therefore, for various needs, the development direction of QKD model is split into a micro level[30,34,35,47,51,52] and a macro level.[53–57] Whether the subjects are system parameters or device parameters is the line between them.

For a micro model, it usually concentrates on the imperfections of devices, such as the non-ideal splitting ratio of beamsplitters, finite extinction ratio of intensity modulators, and channel noise. Such a model is used to solve the engineering problem of hardware systems, which cannot usually be abstracted as an analytic formula due to the complexity caused by the combination of massive imperfections. Therefore, the micro model is essential to model the devices and the connection among the devices. The simulation of the model is usually numerically performed in a dynamic simulation environment, such as OMNet++.[58–60]

Polymerizing the detailed and massive devices' imperfections to a few abstracted system parameters can build a macro model. For instance, the misalignment error[30,52,61,62] is a comprehensive index of imperfections such as the phase drift and finite extinction ratio. Such a model can provide an analytic solution to the QKD systems, which is helpful for the security analysis and parameter optimization, meanwhile, the price is the loss of details.

These two ideas of modeling are the products of different needs, more details and vividness is the target of the micro models, and abstracting macro parameters so that they can cover the imperfections of practical devices is the trend towards the macro models. Although the micro model can theoretically implement the function of the macro model and takes great flexibility, it needs significant computation power to achieve that. Therefore, the line between them may not be blurred in a little time.

3.2.2. Method

The optimization of the secure key rate needs an abstract model that can connect the systems parameters and the gains and quantum bit error rates (QBERs). Here, we abstract the QKD system to three parts, the source (a phase-randomized pulsed laser source), the operation to the quantum state (encoder, channel and decoder), and the detection unit (detectors).

Source: The laser source in QKD is pulsed as the system clock. The implementation of decoy method requires the source is a mixed state of photon-number state, which means the phase of WCPs is randomized.

$$\rho = \int_0^{2\pi} \frac{\mathrm{d}\theta}{2\pi} |\alpha\rangle\langle\alpha| = \sum_n \frac{e^{-\mu}\mu^n}{n!} |n\rangle\langle n| \tag{1}$$

where $\mu = |\alpha^2|$ is the mean photon number of the pulses.

In practical QKD systems, the pulsed laser source can be realized by trigging the pulsed laser or chopping the continuous-wave laser. The former can naturally satisfy the phase randomness, the latter may need active phase randomization according to the relationship between the coherence time and repetition period. The advantage of the chopped continuous-wave laser is the narrow line-width which is useful in measurement-device-independent and twin-field systems. Therefore, the phase randomization is an extra burden in such implements. Recently, the security analysis without phase randomization is proposed[63] that appears to solve the problem. Moreover, the intensity fluctuations of the source are also analyzed.[34,52]

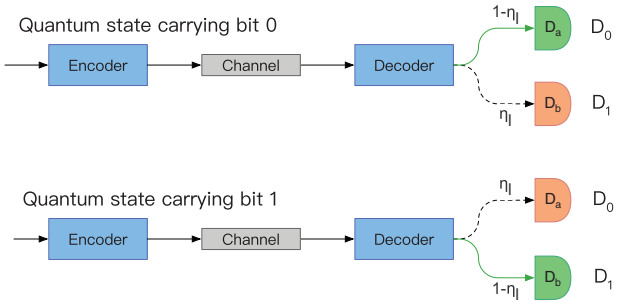


Figure 3. Schematic diagram of encoding and decoding a quantum state with the dual-detector unit. The encoder and decoder prepare quantum states according to key bits and basis bits and map them to a different path based on basis bits, respectively, which consisted of optical components. The response of different detector represents the different key extracted. The η_l describes the probability that a single photon is mapped to an inappropriate path, thereby introducing the misalignment error.

Encoder, Channel, and Decoder: The encoder is used to load classic information onto the quantum state, then the channel performs a transformation to it and the encoder maps the quantum state to the path state for the detection using single-photon detectors. Ideally, apart from the channel, the encoder and decoder with perfect independent state preparation and measurement would not introduce error responses. However, the imperfections of coders are inevitable. For example, the patterning effect[64,65] can bias the intensity of decoy states and the imperfect state preparation and measurement can introduce misalignment errors. Therefore, there are various efforts to build models to describe and deal with the imperfections.[47]

Here we care about the calculation of the gain and QBER. Because they are in direct relations to the misalignment error, which is a comprehensive reflection of the imperfections of above devices, we adopt the misalignment model proposed in ref. [52] to derive the quantum state before the detection.

Specifically, **Figure 3** shows the schematic of encoding and decoding a quantum state in the two-dimensional Hilbert space for the dual-detector unit. The encoder modulates the pulse into the eigenstate of Pauli operators according to the basis and key. The operation of the channel to the quantum state can also be regarded as the Pauli rotation and attenuation. Then, when a light pulse arrives, the decoder can map it to the corresponding path state, so the information carried by the quantum state can be extracted by the responses of the detectors in different paths.

For example, in the phase-encoding system, the quantum state carries the classical bit information vie the phase difference between adjacent laser pulses, so the encoder and decoder can be an asymmetric Faraday–Michelson interferometer. According to

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the phase difference, the decoder can guide the quantum state to a different detector by the interference phenomenon. Then the bit information can be obtained by checking which detector is clicked.

However, all the inaccuracy and instability of encoding and decoding and the disturbance of channel can lead to incorrect guidance of the laser pulse, which can cause an error responds of the detector unit so that extract a flipped bit. Moreover, the multi-photon pulses could even hit the two detectors simultaneously, the consequent possible double-click events can generate error keys with the probability of 50%. These errors are the misalignment errors.

The model of the misalignment error regards the decoder as a broad beamsplitter whose splitting ratio of the intensity of the error pulse to the intensity of the correct pulse can describe the probability of the misalignment error. For example, if the intensity of the incoming pulse is $\mu\eta$, where μ is the initial intensity entering the channel and the η is the overall transmittance including the channel loss and the transmittance of Bob's side, the intensities of two outputs are $\mu\eta(1-\eta_l)$ and $\mu\eta\eta_l$, respectively. When the laser pulse is a single-photon pulse, the η_l degrade into the original misalignment error probability, e_d.

After obtaining the pulse intensity of each output path, the next is to derive the response probabilities of detectors.

Detector: The detection of the single-photon is a key step of measuring quantum states to generate keys.[66,67] For different wavelengths of photon and requirements of performance, the methods to detect photons are diverse. Here, we think about the wavelength of 1550 nm, which is a communication window in optical fibres. Because the energy of a single-photon is too weak, the detector should be sensitive enough to the response of the single-photon or can amplify the weak response signal to a strong one for detection. To detect a photon with the wavelength of 1550 nm, the main methods in QKD are the superconducting nanowire single-photon detector (SNSPD)[68–70] and the single-photon avalanche diode (SPAD),[71–73] which correspond to sensitive detection and amplified detection, respectively.

The SNSPD can produce an electrical signal when an incoming photon is absorbed by the nanowire and subsequently quenches it. It has a satisfactory performance on quantum efficiency and dark counts. However, for practical QKD systems, the volume, cost, and stability need to be accounted for. Therefore, although the detection efficiency of SNSPD is very high, the complex refrigerating system, high price and possible quenching of superconducting state limit its implementation in practical systems.

The other solution is the SPAD. In the SPAD, the absorption of a single photon can excite a carrier, applying a bias field, which can be amplified by avalanche gain. The final avalanche current is so strong that it can be captured by the peripheral circuit. The probability that an incident photon can ignite a detectable avalanche current successfully is the detection efficiency.

The materials of SPAD depend on the wavelength of the photon, whose bandgap should match the energy of the photon. For example, InGaAs/InP is used to grow the SPAD. Unfortunately, there are always some defects formed in the fabrication of diodes. Such defects can trap the avalanche carriers and randomly release them later, which can ignite a new avalanche even no photon arrives. These avalanche response is called the afterpulse.[74–76] To quantitatively measure it, here we define the afterpulse rate as the average number of the afterpulses ignited by an avalanche.

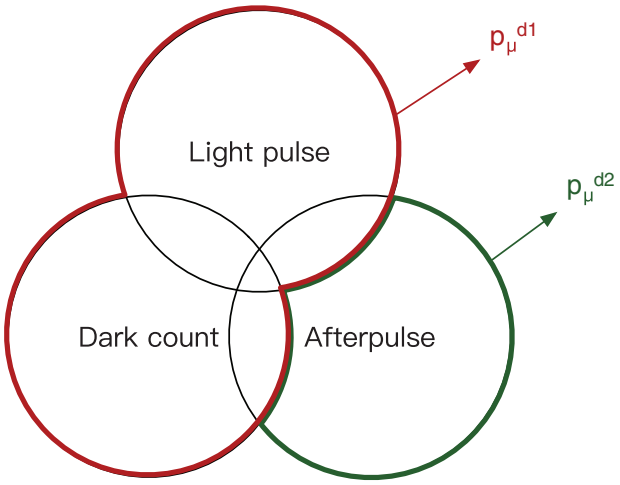


Figure 4. The Venn diagram of the three sources that can ignite the SPAD. p_μ^{a1} represents the original detector model which only considers the click ignited by light pulses and dark counts. p_μ^{a2} is the supplementary part representing the contribution of afterpulses. μ is the mean photon number of the light pulses, and $d \in \{a, b\}$ represents one detector.

Moreover, the primary dark current in SPAD can also ignite an avalanche as a seed carrier.[74] Such an undesirable response is called the dark count. The dark count is usually measured by the probability of detecting a dark count in a detection event or a unit time, which is called dark-count probability.

Therefore, there are three types of events, the light pulse, dark count, and afterpulse, that can produce a response as shown in **Figure 4**.

For detector D_a, the probability of responding to the light pulse and dark count is denoted by $p_\mu^{a_1}$, where μ is the intensity of the pulse hit D_a. $p_\mu^{a_2}$ represents the complementary probability of afterpulse event. Then the overall response probability of D_a is $p_\mu^{a_1} + p_\mu^{a_2}$, where

$$\begin{aligned} p_\mu^{a} &= p_\mu^{a_1} + p_\mu^{a_2} \\ p_\mu^{a_1} &= 1 - e^{-\mu\eta}(1-p_d) \\ p_\mu^{a_2} &= (1-p_\mu^{a_1})P_{ap} \end{aligned} \tag{2}$$

For the response probability of detector D_b, p_μ^b, the results are similar to above equations.

The afterpulse probability P_{ap} depends on the afterpulse rate p_a and the avalanche probability, Q^d. That is

$$P_{ap} = p_a Q^d \tag{3}$$

However, the avalanche can also be an afterpulse. Here, the afterpulse ignited by another afterpulse is called high-order afterpulse. The high-order afterpulse changes the term of p_a to $p_a/(1-p_a)$.[51,77] Then the Q^d is changed to $\hat{Q}^d$ which only contains the events of light pulse and dark count. In decoy-state

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protocols, it is a weighted average of the response probability of different decoy states,

$$\hat{Q}^d = \sum_{\alpha=\mu_1,\mu_2,\ldots,\mu_n} P_\alpha \widetilde{Q}_\alpha \tag{4}$$

where $\widetilde{Q}_\alpha$ is the response probability of μ_n state.

Combined with the misalignment model, the intensity of the pulses hit D_a and D_b are $\mu(1-\eta_l)$ and $\mu\eta_l$, respectively. So the p_μ^a and p_μ^b can be given by

$$\begin{aligned} p_\mu^a &= 1 - e^{-\mu\eta(1-\eta_l)}(1-p_d)(1-P_{ap}) \\ p_\mu^b &= 1 - e^{-\mu\eta\eta_l}(1-p_d)(1-P_{ap}) \end{aligned} \tag{5}$$

where

$$\begin{aligned} P_{ap} &= \frac{p_a}{1-p_a}\hat{Q}^d \\ \hat{Q}^d &= \sum_{\alpha=\mu_1,\mu_2,\ldots,\mu_n} P_\alpha \widetilde{Q}_\alpha \\ \widetilde{Q}_\alpha &= (P_X^2 + P_Z^2)(1 - \frac{1}{2}(e^{-\alpha\eta(1-\eta_l)} + e^{-\alpha\eta\eta_l})(1-p_d)) \\ &\quad + 2P_X P_Z(1 - e^{-\frac{\alpha\eta}{2}}(1-p_d)) \end{aligned} \tag{6}$$

where $\widetilde{Q}_\alpha$ is a weighted average of the response probabilities of successful and failed basis sifting.

Finally, according to the definition of the valid response in Section 2, the probability of getting a successful response and a error response are respectively given by

$$\begin{aligned} Q_\mu &= p_\mu^a(1-p_\mu^b) + (1-p_\mu^a)p_\mu^b + p_\mu^a p_\mu^b \\ T_\mu &= (1-p_\mu^a)p_\mu^b + \frac{1}{2}p_\mu^a p_\mu^b \end{aligned} \tag{7}$$

where Q_μ and $T_\mu = E_\mu Q_\mu$ are overall yield and error yield of μ state.

3.3. Calculation

The theoretical security of QKD has been proofed by the uncertainty principle, the entanglement distillation protocol and the information theory. However, to generate the secure key in a practical system need to consider the imperfections of devices and the limited resource, that is, the finite-key effect.

Decoy Method for the Source Containing Multi-Photon Pulses: GLLP formula[22] builds the framework of distilling the secure key with imperfection devices. It indicates the single-photon pulses are the source of secure keys and secures the QKD using the non-ideal single-photon source systems by the worst estimations of single-photon parameters, that is, the lower bounds of the yields of single-photon events and vacuum-state events and the upper bound of the phase-error yield of single-photon events. The former and the latter are used to estimate the mutual information between Alice and Bob, Bob and Eve, respectively. For example, the lower bound of single-photon yield, Y_1^-, and the upper bound of the phase-error yield, $e_{1,p}^+$, are, respectively, given by

$$\begin{aligned} Y_1^- &= \frac{Q_\mu - P_{multi}}{Q_\mu} \\ e_{1,p}^+ &= e_1^+ = \frac{E_\mu}{Y_1^-} \end{aligned} \tag{8}$$

in the asymptotic scenario where the phase-error yield, $e_{1,p}$, is equal to the bit-error yield, e_1, the superscripts of + and − are represents the upper and lower bounds, respectively.

However, such estimations are not tight enough which greatly limits the secure communication distance. The decoy-state method can provide a tight estimation by randomly preparing decoy states with various intensities. Then the parameters can be obtained by solving the following problems

$$\begin{aligned} &\min Y_1, Y_0; \max e_1 \\ s.t.\quad & Q_{\mu_1} = \sum_{i=0}^{\infty} p_{i|\mu_1} Y_i \\ & T_{\mu_1} = \sum_{i=0}^{\infty} p_{i|\mu_1} Y_i e_i \\ & \cdots \\ & Q_{\mu_n} = \sum_{i=0}^{\infty} p_{i|\mu_n} Y_i \\ & T_{\mu_1} = \sum_{i=0}^{\infty} p_{i|\mu_n} Y_i e_i \end{aligned} \tag{9}$$

where the $p_{i|\mu_1}$ is the photon-number probability and, for example, follows the Poisson distribution when the laser source is a phase-randomized weak coherent source and other special probability distributions in passive decoy-state schemes.[36,78–85]

Based on this idea, researchers have developed various methods to obtain tighter bounds. Here, we adopt the method proposed in ref. [37] which can tighten the bounds by introducing an additional condition,

$$\begin{aligned} Y_1^{Z,Z} &= Y_1^{Z,X} \\ Y_1^{Z,X} &= Y_1^{X,X} \end{aligned} \tag{10}$$

With these constraint conditions, the bounds of Y_1 and e_1 can be derived by mathematical programming or analytic formulas, the detailed analytic bounds are provided in Appendix A.

Regulation for the Finite-Key Effect: The finite-key effect introduces the deviation between the probability and the frequency of yields and the deviation between the phase error and the bit error. Such problems can be dealt with the idea of random sampling without replacement. For the former, the observables of yields are related to their expectations by Hoeffding's inequality,[86] one of the concentration inequalities. Specifically, for the observed number n of an event, Hoeffding's inequality can provide a confidence interval of its expectation value, $[n^-, n^+]$, with a fiducial probability of $1 - 2\tilde{\varepsilon}_{sec}$. With this tool, the upper bounds and lower bounds of overall yield and error yield are obtained, and

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the equality constraints in Equation (9) can be replaced with inequality constraints like $Q^{+}_{\mu_1} \leq \sum_{i=0}^{\infty} p_{i|\mu_1} Y_i \leq Q^{-}_{\mu_1}$.[87] The detailed expressions of the bounds are shown in Appendix A.

On the other hand, the phase-error rate cannot be replaced by the bit-error rate in the finite-size scenario. For two mutually unbiased bases, X and Z basis, for example, the transformation of the untrusted channel that causes bit errors in one basis presents phase errors in the other basis. Refs. [39, 42] ingeniously adopt the hypergeometric distribution to bound the deviation between them. Specifically, when the length of the sifted key is limited to a finite number, N, and they are supposed to be measured in Z basis, the total number of phase error events is assumed as m. Then, to bound the number of phase events in Z basis, n_x events are selected from N and measured in X basis, thus the original phase errors in n_x events show up as bit errors measuring with X basis. The number of these bit errors in X basis follows the hypergeometric distribution of $H(n, M, N)$. Then based on the probability distribution, the upper bound of the phase-error yield in Z basis can be derived as $e^{Z}_{1,p} \leq e^{X}_{1} + g$ with a fiducial probability of $1 - \tilde{\varepsilon}_{sec}$. The detailed expressions of the bound are also shown in Appendix A.

Finally, combined with the composable security analysis, the length of the secure key, l_ω, obtained in ω basis and given by[41]

$$l_\omega = s_0^{\omega-} + s_1^{\omega-}\left[1 - H_2\left(e_{1,p}^{\omega+}\right)\right] - \lambda_{EC} - 6\log_2\frac{1}{\tilde{\varepsilon}_{sec}} - \log_2\frac{2}{\varepsilon_{cor}} \quad (11)$$

where s_0 is the number of vacuum events, s_1 is the number of single-photon events, $e_{1,p}$ is the phase error rate, the superscripts, + and − represent upper and lower bounds, respectively, $H_2(x) = -x\log_2(x) - (1-x)\log_2(1-x)$ is the binary Shannon entropy function, $\lambda_{EC} = n^{\omega} f_e H_2(E_\omega)$ is the consumption of the information in error-correction, f_e is the efficiency factor of the error-correction method used, ε_{cor} and $\tilde{\varepsilon}_{sec}$ are secure parameters, and $\tilde{\varepsilon}_{sec}$=$\varepsilon_{sec}$/21 which is the most pessimistic scenario.

Then, the total secure key rate R can be generated by X basis and Z basis.

$$R = \frac{\sum_\omega l_\omega}{N} \quad (12)$$

3.4. Optimization

Based on the statement above, the secure key rate can be written as the following formula

$$R = f(I, P, A_{sys}) \quad (13)$$

where $I = \{\mu_{Z1}, \ldots, \mu_{Zn}, \mu_{X1}, \ldots, \mu_{Xm}\}$ is the set of the intensities of weak coherent pulses which depends on the particular scheme of decoy, $P = \{p_{\mu_{Z1}}, \ldots, p_{\mu_{Zn}}, p_{\mu_{X1}}, \ldots, p_{\mu_{Xm}}, p_X, p_Z\}$ is the set including the probabilities that Alice prepares the pulse with particular intensity and basis and the basis-choice probabilities of Bob, and the system parameters are included in A_{sys} and listed in Table 2.

According to the given system parameters A_{sys}, I and P need to be optimized for the optimal secure key rate, which can be abstracted into an optimization problem,

$$\begin{aligned} \max \quad & R(I, P, A_{sys}) \\ s.t. \quad & I \geq 0 \\ & 0 \leq P \leq 1 \\ & \sum P = 1 \end{aligned} \quad (14)$$

where the objective function of the problem is the formula of the secure key rate, and the decision variables represent the system parameters related to the secure key rate.

To solve the optimization problem, the local search algorithm (LSA) is a common option due to the convexity of the secure key rate function.[88] However, its time complexity increases with the number of decision variables exponentially. Moreover, it might not be a convexity function when the key rate is generated by both Z basis and X basis, which can converge the LSA on a local optimum. Therefore, here we employ the particle swarm optimization (PSO),[89] a global optimization algorithm, to avoid potential trouble.

The PSO first generates several trials in the solution space. Then, they continuously update their position in the space according to the personal best, the global best, and a random disturbance. When the algorithm converged, all trials are collected at the globally optimal solution, also the personal best position.

Specifically, updating mechanism for the trial $P_{old} = \{x_1, x_2, \ldots, x_n\}$ is

$$\begin{aligned} P_{new} &= P_{old} + V_{new} \\ V_{new} &= wV_{old} + C_p r_p(P_p - P_{old}) + C_g r_g(P_g - P_{old}) \end{aligned} \quad (15)$$

where V is the velocity vector represented the Euclidean distance between the old and new position, P_p and P_g are the personal best and global best, respectively, w, C_p, and C_g are the hyper-parameters of the algorithm, r_p and r_g are the binary random number.

The search space of PSO does not change over the number of decision variables. Generally speaking, the computation time of PSO is longer than LSA if there are few decision variables. However, with the increasing of the search space, LSA may not be able to converge in a countable time. For example, in some schemes, the number of decision variables is more than ten, which cannot be solved by LSA. For a fair comparison, PSO is used for all decoy-state protocols in the numerical simulation.

4. Results and Discussion

In this section, we simulate 38 decoy-state protocols whose total number of intensity, QTY, is from two to six, as shown in **Table 1**. Specifically, the scheme quantities of 2-intensity, 3-intensity, 4-intensity, 5-intensity, and 6-intensity decoy method are 1, 9, 22, 5, and 1, respectively.

Here, we study the optimal decoy-state protocol with five factors: transmission loss, block size, misalignment-error probability, afterpulse rate, and dark-count probability, corresponding to η_c, n^Z, η_l, p_a, and p_d, respectively. In the simulations, all the system parameters are valued as **Table 2** except for the investigative

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Table 1. Decoy configurations from 2-intensity scheme to 6-intensity schemes.

QTY	N_{IZ}	*Xlock*	Configuration	QTY	N_{IZ}	*Xlock*	Configuration
2	2	{1, 2}	$\mu_{Z1}=\mu_{X1}, \mu_{Z2}=\mu_{X2}$	4	3	{0, 3}	$\mu_{Z1}, \mu_{X1}, \mu_{Z2}, \mu_{Z3}=\mu_{X2}$
3	2	{0, 2}	$\mu_{Z1}, \mu_{X1}, \mu_{Z2}=\mu_{X2}$	4	3	{1, 0}	$\mu_{Z1}=\mu_{X1}, \mu_{Z2}, \mu_{Z3}$
3	2	{1, 0}	$\mu_{Z1}=\mu_{X1}, \mu_{Z2}, \mu_{X2}$	4	3	{2, 0}	$\mu_{Z1}, \mu_{Z2}=\mu_{X1}, \mu_{Z3}$
3	2	{0, 1, 2}	$\mu_{X1}, \mu_{Z1}=\mu_{X2}, \mu_{Z2}=\mu_{X3}$	4	3	{3, 0}	$\mu_{Z1}, \mu_{Z2}, \mu_{Z3}=\mu_{X1}$
3	2	{1, 2, 0}	$\mu_{Z1}=\mu_{X1}, \mu_{Z2}=\mu_{X2}, \mu_{X3}$	4	3	{0, 2, 3}	$\mu_{Z1}, \mu_{X1}, \mu_{Z2}=\mu_{X2}, \mu_{Z3}=\mu_{X3}$
3	2	{1, 0, 2}	$\mu_{Z1}=\mu_{X1}, \mu_{X2}, \mu_{Z2}=\mu_{X3}$	4	3	{2, 0, 3}	$\mu_{Z1}, \mu_{Z2}=\mu_{X1}, \mu_{X2}, \mu_{Z3}=\mu_{X3}$
3	3	{1, 2}	$\mu_{Z1}=\mu_{X1}, \mu_{Z2}=\mu_{X2}, \mu_{Z3}$	4	3	{2, 3, 0}	$\mu_{Z1}, \mu_{Z2}=\mu_{X1}, \mu_{Z3}=\mu_{X2}, \mu_{X3}$
3	3	{2, 3}	$\mu_{Z1}, \mu_{Z2}=\mu_{X1}, \mu_{Z3}=\mu_{X2}$	4	3	{1, 0, 3}	$\mu_{Z1}=\mu_{X1}, \mu_{X2}, \mu_{Z2}, \mu_{Z3}=\mu_{X3}$
3	3	{1, 3}	$\mu_{Z1}=\mu_{X1}, \mu_{Z2}, \mu_{Z3}=\mu_{X2}$	4	3	{1, 3, 0}	$\mu_{Z1}=\mu_{X1}, \mu_{Z2}, \mu_{Z3}=\mu_{X2}, \mu_{X3}$
3	3	{1, 2, 3}	$\mu_{Z1}=\mu_{X1}, \mu_{Z2}=\mu_{X2}, \mu_{Z3}=\mu_{X3}$	4	3	{0, 1, 3}	$\mu_{X1}, \mu_{Z1}=\mu_{X2}, \mu_{Z2}, \mu_{Z3}=\mu_{X3}$
4	2	{0, 0}	$\mu_{Z1}, \mu_{X1}, \mu_{Z2}, \mu_{X2}$	4	3	{1, 2, 0}	$\mu_{Z1}=\mu_{X1}, \mu_{Z2}=\mu_{X2}, \mu_{Z3}, \mu_{X3}$
4	2	{0, 0, 2}	$\mu_{Z1}, \mu_{X1}, \mu_{X2}, \mu_{Z2}=\mu_{X3}$	4	3	{1, 0, 2}	$\mu_{Z1}=\mu_{X1}, \mu_{X2}, \mu_{Z2}=\mu_{X3}, \mu_{Z3}$
4	2	{0, 1, 0}	$\mu_{X1}, \mu_{Z1}=\mu_{X2}, \mu_{Z2}, \mu_{X3}$	4	3	{0, 1, 2}	$\mu_{X1}, \mu_{Z1}=\mu_{X2}, \mu_{Z2}=\mu_{X3}, \mu_{Z3}$
4	2	{0, 2, 0}	$\mu_{Z1}, \mu_{X1}, \mu_{Z2}=\mu_{X2}, \mu_{X3}$	5	2	{0, 0, 0}	$\mu_{Z1}, \mu_{Z2}, \mu_{X1}, \mu_{X2}, \mu_{X3}$
4	2	{1, 0, 0}	$\mu_{Z1}=\mu_{X1}, \mu_{Z2}, \mu_{X2}, \mu_{X3}$	5	3	{0, 0}	$\mu_{Z1}, \mu_{X1}, \mu_{Z2}, \mu_{X2}, \mu_{Z3}$
4	2	{2, 0, 0}	$\mu_{Z1}, \mu_{Z2}=\mu_{X1}, \mu_{X2}, \mu_{X3}$	5	3	{0, 0, 3}	$\mu_{Z1}, \mu_{X1}, \mu_{Z2}, \mu_{X2}, \mu_{Z3}=\mu_{X3}$
4	2	{0, 0, 1}	$\mu_{Z1}=\mu_{X3}, \mu_{X1}, \mu_{Z2}, \mu_{X2}$	5	3	{0, 2, 0}	$\mu_{Z1}, \mu_{X1}, \mu_{Z2}=\mu_{X2}, \mu_{Z3}, \mu_{X3}$
4	3	{0, 1}	$\mu_{Z1}=\mu_{X2}, \mu_{Z2}, \mu_{Z3}$	5	3	{1, 0, 0}	$\mu_{Z1}=\mu_{X1}, \mu_{Z2}, \mu_{X2}, \mu_{Z3}, \mu_{X3}$
4	3	{0, 2}	$\mu_{Z1}, \mu_{X1}, \mu_{Z2}=\mu_{X2}, \mu_{Z3}$	6	3	{0, 0, 0}	$\mu_{Z1}, \mu_{Z2}, \mu_{Z3}, \mu_{X1}, \mu_{X2}, \mu_{X3}$

Table 2. List of the system parameters in A_{sys} and used in numerical simulations.

n^Z	f_e	ε_{sec}	ε_{cor}	η_I	η_{Bob}	p_a	p_d	η_c
10^9	1.16	10^{-9}	10^{-15}	3.3%	10%	5%	6×10^{-7}%	10 dB

n^Z is the total number of responses on Z basis. f_e is the error correction efficiency. ε_{sec} is the security parameters of finite key. ε_{cor} is the probability that a pair of nonidentical keys passes the error-verification step. η_I is the probability of misalignment error. η_{Bob} is the total transmittance of Bob's side including the detector efficiency. p_a and p_d are the afterpulse rate and the dark-count probability of detectors. η_c is the typical fiber channel loss of 50 km

factor. These factors cover most systems with various characteristics. So based on these results, the optimal decoy scheme in different scenarios can be solved as a reference for practical systems.

In the following sections, the optimal decoy-state protocol in different scenarios is provided and discussed. Moreover, the secure key rates of all 38 decoy-state protocols are shown in the Appendix in detail.

4.1. SKR versus Transmission Loss

We first provide the simulation results of the optimal decoy-state protocol as a function of the transmission loss. In the simulation, n^Z is required to be larger than n^X to avoid the bug that the overall yield of Z basis can be extremely low. That happens because the block size of postprocessing in the finite-key scenario is n^Z and fixed to a constant value, if n^Z is not limited to be larger than n^X, the optimization algorithm can be trapped by the solution that the preparing probability and intensity of Z-basis states are extremely low, which leads to a near asymptotic key rate in X basis. However, it is useless in practical systems.

Figure 5 shows the optimal decoy-state protocol over the transmission loss. Considering the number of decoy state can be limited, we also compare them for the optimal decoy-state schemes of 3-intensity, 4-intensity, and 5-intensity protocols. The 2-intensity and 6-intensity protocols are excepted from the internal comparison because they have only one decoy scheme.

The results show that the optimal scheme is the 2-000 for all transmission losses, where Alice independently prepares two intensities of decoy states in Z basis and three intensities of decoy states in X basis. The optimal schemes, 2-012 and 2-002, for 3-intensity protocol and 4-intensity protocol represent that, for 3-intensity protocol (4-intensity protocol), Alice independently modulates two intensities of decoy states in Z basis and one (two) intensities of decoy states in X basis, the intensities of the rest states of X basis are equal to the intensities in Z basis.

Specifically, for the 2-000 protocol, Alice prepares five decoy states, μ_{Z1}, μ_{Z2}, μ_{X1}, μ_{X2}, μ_{X3}, with probabilities $p_{\mu_{Z1}}$, $p_{\mu_{Z2}}$, $p_{\mu_{X1}}$, $p_{\mu_{X2}}$, $p_{\mu_{X3}}$; for the 2-012 protocol, Alice prepares five decoy states, μ_{Z1}, μ_{Z2}, μ_{X1}, $\mu_{X2}\equiv\mu_{Z1}$, $\mu_{X3}\equiv\mu_{Z2}$, with probabilities $p_{\mu_{Z1}}$, $p_{\mu_{Z2}}$, $p_{\mu_{X1}}$, $p_{\mu_{X2}}$, $p_{\mu_{X3}}$; for the 2-000 protocol, Alice prepares five decoy states, μ_{Z1}, μ_{Z2}, μ_{X1}, $\mu_{X2}\equiv\mu_{Z2}$, μ_{X3}, with probabilities $p_{\mu_{Z1}}$, $p_{\mu_{Z2}}$, $p_{\mu_{X1}}$, $p_{\mu_{X2}}$, $p_{\mu_{X3}}$,

In these scenarios, the optimal schemes do not change. In fact, with the increasing of transmission loss, the valid counts should have decreased, which can magnify the finite-size effect and influence the performance of decoy-state protocols with different quantity of states. However, using the finite-size analysis based on the sampling without replacement, the valid counts are accumulated to a fixed value, n^Z, which abates the effect of changing the transmission loss.

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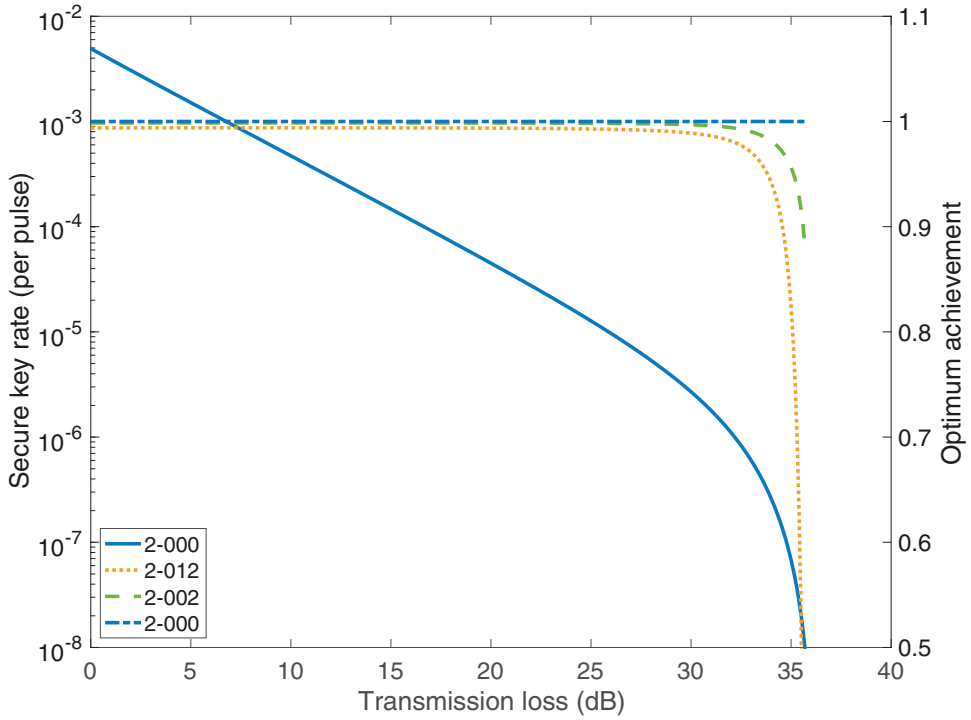


Figure 5. The optimal decoy-state protocols as functions of the transmission loss. The solid line is the optimal secure key rate in all protocols of decoy state corresponding to the left *y*-axis. The dotted line, dashed line, and dash-dot line are the achievements versus the optimal secure key rate for 3-intensity, 4-intensity, and 5-intensity encoding systems, respectively. Specifically, the achievement is the ratio of the secure key rate to the optimal one. The colors in lines represent decoy-state protocols shown in the legend.

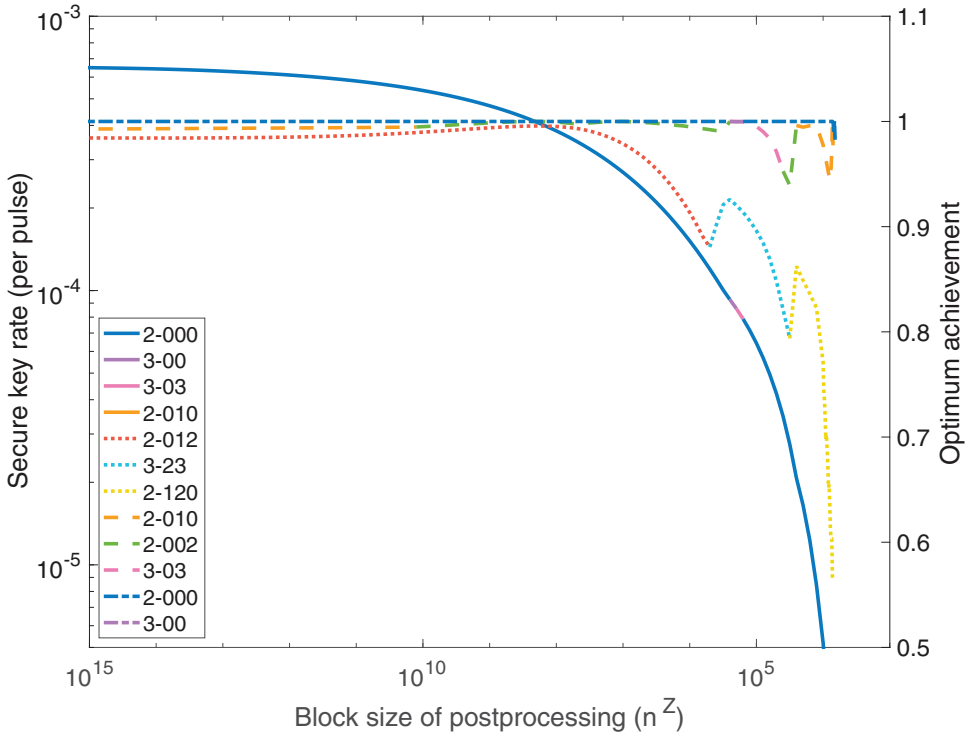


Figure 6. The optimal decoy-state protocols as functions of the block size of postprocessing. The solid line is the optimal secure key rate in all protocols of decoy state corresponding to the left *y*-axis. The dotted line, dashed line, and dash-dot line are the achievements versus the optimal secure key rate for 3-intensity, 4-intensity, and 5-intensity encoding systems, respectively. Specifically, the achievement is the ratio of the secure key rate to the optimal one. The colors in lines represent decoy-state protocols shown in the legend.

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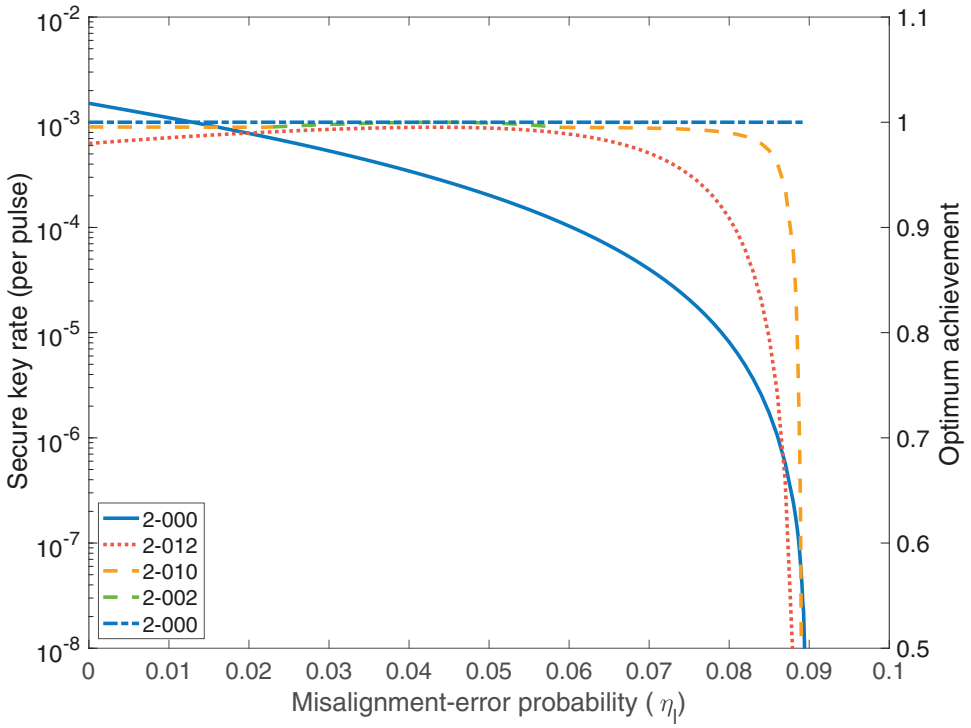


Figure 7. The optimal decoy-state protocols as functions of the probability of misalignment error. The solid line is the optimal secure key rate in all protocols of decoy state corresponding to the left *y*-axis. The dotted line, dashed line, and dash-dot line are the achievements versus the optimal secure key rate for 3-intensity, 4-intensity, and 5-intensity encoding systems, respectively. Specifically, the achievement is the ratio of the secure key rate to the optimal one. The colors in lines represent decoy-state protocols shown in the legend.

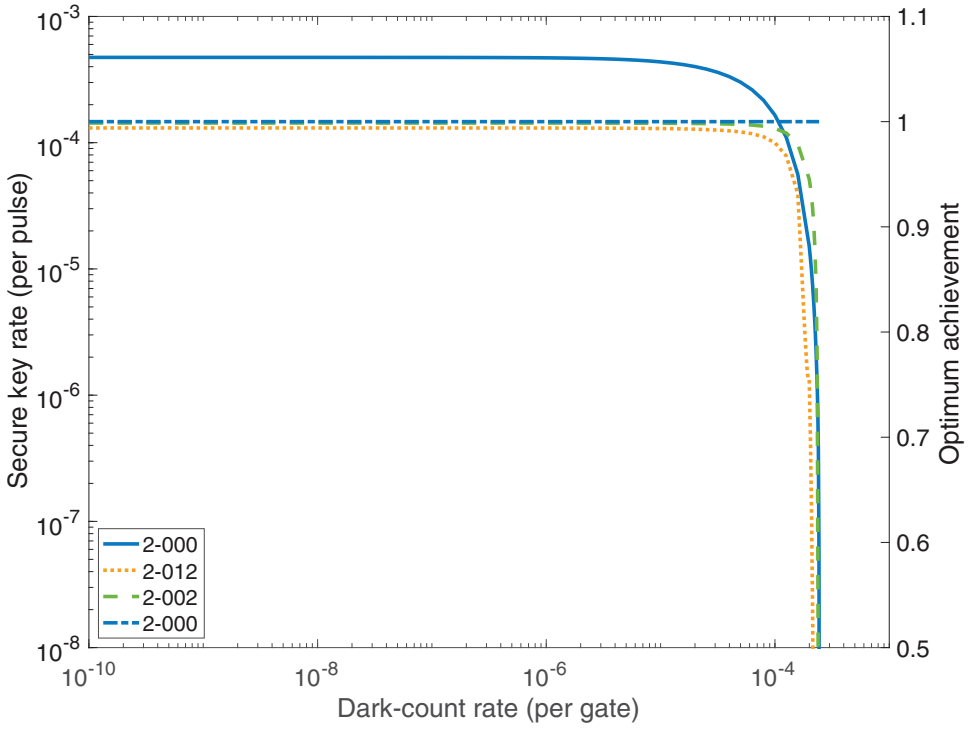


Figure 8. The optimal decoy-state protocols as functions of the dark-count probability. The solid line is the optimal secure key rate in all protocols of decoy state corresponding to the left *y*-axis. The dotted line, dashed line, and dash-dot line are the achievements versus the optimal secure key rate for 3-intensity, 4-intensity, and 5-intensity encoding systems, respectively. Specifically, the achievement is the ratio of the secure key rate to the optimal one. The colors in lines represent decoy-state protocols shown in the legend.

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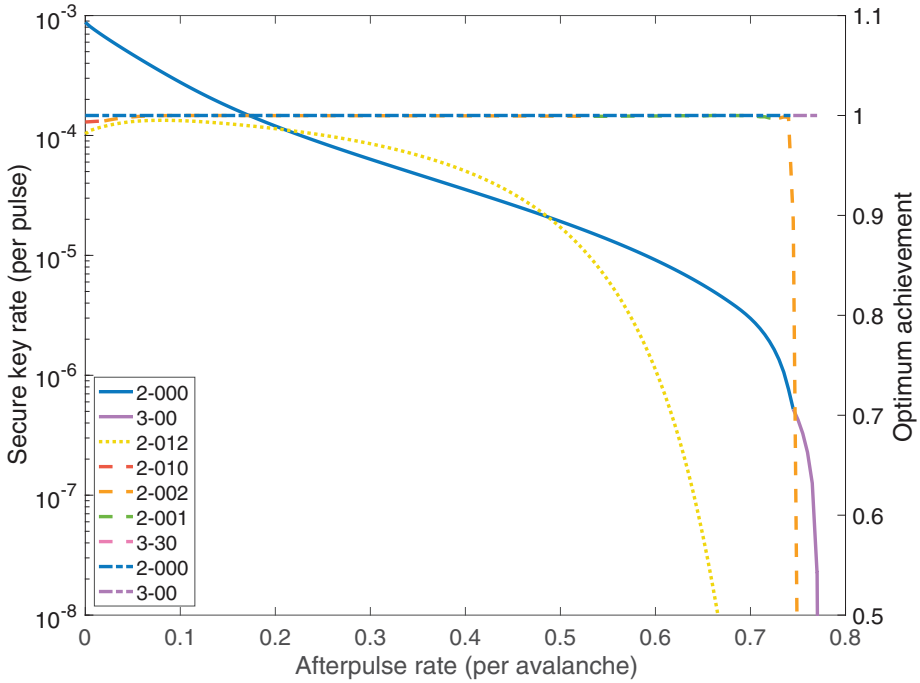


Figure 9. The optimal decoy-state protocols as functions of the afterpulse rate. The solid line is the optimal secure key rate in all protocols of decoy state corresponding to the left *y*-axis. The dotted line, dashed line, and dash-dot line are the achievements versus the optimal secure key rate for 3-intensity, 4-intensity, and 5-intensity encoding systems, respectively. Specifically, the achievement is the ratio of the secure key rate to the optimal one. The colors in lines represent decoy-state protocols shown in the legend.

4.2. SKR versus Block Size

In this section, we study the optimal decoy scheme with changing block size of postprocessing, n^Z, at the transmission loss of 10 dB, ≈50 km of standard fiber channel. The QKD system can dynamically regulate n^Z for different implement scenarios. For example, a fewer n^Z can shorten the waiting time of postprocessing and allows the systems to generate keys with a higher frequency, which is valuable to short massage communication because the user prefers to wait one second for five thousand bits of key rather than wait for one thousand seconds for fifty thousand bits of the key.

Changing n^Z can bring significant finite-size effect to the secure key rate, thus the optimal decoy-state protocol is not changeless which differs from Figure 5. For all schemes, there are four protocols occupying the optimal one at different block size shown as the solid line in Figure 5, which belong to 4-intensity (3-03 and 2-010) and 5-intensity (2-000 and 3-00) protocols.

Moreover, other lines in **Figure 6** show the change of optimal protocol in specific systems. For example, for 3-intensity systems, when n^Z is larger than 10^6, the secure key mainly comes from Z basis, and the dominant protocol is 2-012. Then, as n^Z decreases, the key-rate contribution of X basis is increasing. For protocol 3-23, more intensities in Z basis can tighten the parameter estimation for X basis, which brings a high secure key rate for X basis, so the dominant protocol is changed to 3-23. Finally, the extremely small n^Z increases the error rate of Z basis causing a slide of the key rate, thus protocol 2-120 becomes the optimum due to the higher yield and lower error yield of single-photon state in Z basis.

4.3. SKR versus Misalignment-Error Probability

The sources of misalignment error are different in the QKD systems with different encoding schemes and physical channels. For example, the finite extinction ratio of the optical switch or polarization beam splitter can cause misalignment errors in time-bin and polarization encoding systems, respectively, and the birefringence of fiber can introduce misalignment errors in phase-encoding systems. Moreover, the channel disturbance can perform a transformation to the quantum states which can lead to misalignment errors. Therefore, the misalignment errors present different characteristics in different systems and are necessary for analysis.

Figure 7 shows the optimal decoy-state protocol as a function of misalignment-error probability. Unlike the aforementioned figure, the optimal protocol only changes in the 4-intensity decoy-state systems. Because the effect of misalignment error on the choice of decoy-state protocols differs from the finite-size effect of transmission loss and block size. The higher error rate caused by the misalignment errors requires more precise estimation of the single-photon error yield, which depends on the data obtained in X basis. Therefore, the number of states in X basis is optimized to three for a tighter estimation.

4.4. SKR versus Dark-Count Probability

The dark-count probability is an important characteristic of detector units, similar to the misalignment-error probability, which can increase the error yield of the QKD systems. Previous works[77] have shown that the optimal dark-count probability can

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be different with various communication distance. Therefore, regulating the dark-count probability of detectors might be a demand for high system performance.

Because the dark-count probability also contributes error counts to error yield like the misalignment-error probability, the results of optimal decoy-state protocol shown in **Figure 8** are similar to Figure 7. Therefore, in all cases, preparing two states in Z basis and three states in X basis is the optimal choice.

4.5. SKR versus Afterpulse Rate

The afterpulse rate, as a characteristic of single-photon avalanche detector, is also the main source of error rate and has a more significant effect on the protocol choice than dark-count probability. The afterpulse probability is related to the afterpulse rate of detectors and the intensity of decoy states, especially the signal state because the main effect of afterpulse effect is the afterpulses ignited by signal state biased the yields of other weaker decoy states.[51] Therefore, the intensities of decoy states would decrease with the increasing afterpulse rate.

Figure 9 shows the optimal decoy-state protocols as functions of the afterpulse rate and reflects the error-rate effect on the choice of decoy-state protocol significantly. Due to the strong influence of error counts, there is a more frequent changing of the optimal decoy-state protocols in Figure 9. Meanwhile, the change of the main basis of key generation is also presented, especially for the change of optimal protocols form 2-000 to 3-00 which are symmetrical protocols corresponding to the dominant basis of Z basis and X basis, respectively.

5. Conclusion

In this paper, we synthesize the recent advances of modeling and optimizing the decoy-state protocol to analyze the optimal one in different scenarios as a reference to practical QKD systems. Specifically, the finite-size analysis based on the sampling without replacement, the model of afterpulse effect and misalignment error and the tighter bound obtaining by mismatch bases is combined in the secure key generation.

We analyze 38 decoy-state protocols using this method with the changes of transmission loss, block size of postprocessing,

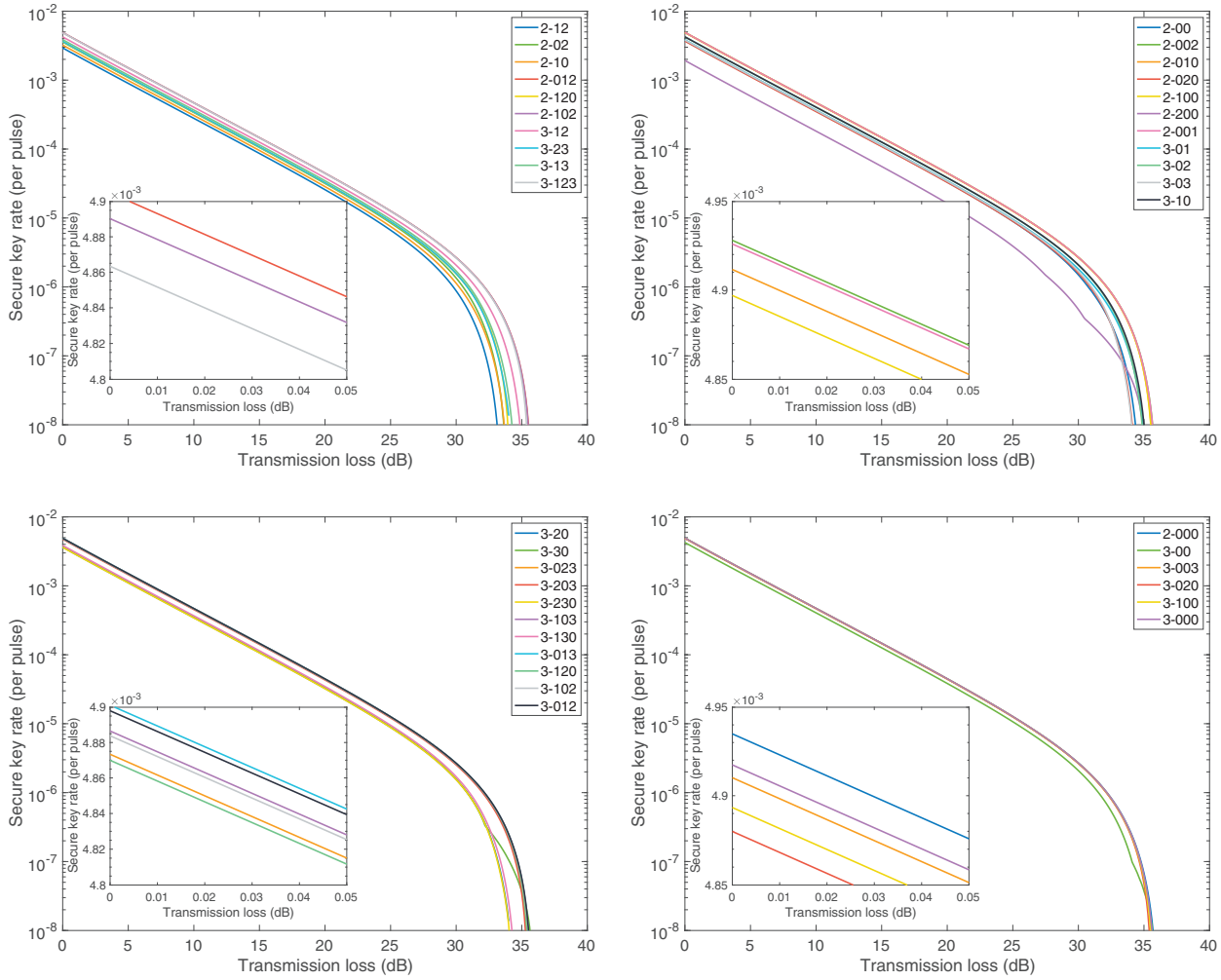


Figure B1. Secure key rates of the decoy-state protocols as functions of the transmission loss listed in Table 1. Each figure has a subfigure to show the magnification of the local area at 0 dB. The digits of the legend located in the left and the right to the hyphen are the N_{IZ} and *Xlock*, respectively.

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misalignment-error probability, dark-count probability, and afterpulse rate, corresponding to different QKD systems and implement scenarios. The optimal decoy-state protocols in each scenario are revealed. Moreover, the effects of these control variables on the choice decoy-state protocols are different. The first two factors introduce finite-size effect to influence the optimal protocol, and the rest prefer more decoy states to obtain a tighter bound of the single-photon error yield and weaken the effect of the error counts they brought.

The software and hardware design of a practical QKD system is critical to obtain a high performance, where the decoy-state protocol in the finite-size scenario is the core factor to determine the optical and electronic scheme and the running-time parameters of the system. The synthesis and analysis of optimizing decoy-state protocol can provide an efficient reference for practical QKD systems to help them in reaching the performance peak in various user scenarios.

Appendix A: Analytic Bounds of Parameters Estimation

Here, when the number of decoy states preparing in ω_a basis is two, the upper bound and lower bound of vacuum-state yields are given by

$$Y_0^{\omega_b,-} = \max\left\{\frac{p_{1|\mu_{\omega_a 2}} Q^{\omega_b}_{\mu_{\omega_a 1}} - p_{1|\mu_{\omega_a 1}} Q^{\omega_b}_{\mu_{\omega_a 2}}}{p_{0|\mu_{\omega_a 1}} p_{1|\mu_{\omega_a 2}} - p_{0|\mu_{\omega_a 2}} p_{1|\mu_{\omega_a 1}}} | \omega_a \in \{X, Z\}\right\}$$

$$Y_0^{\omega_b,+} = \min\left\{\frac{2T^{\omega_b}_{\mu_{\omega_a i}}}{p_{0|\mu_{\omega_a i}}}, \frac{Q^{\omega_b}_{\mu_{\omega_a i}}}{p_{0|\mu_{\omega_a i}}} | \mu_{\omega_a i} \in \{\mu_{Z1}, \ldots, \mu_{Zn}, \mu_{X1}, \ldots, \mu_{Xm}\}\right\} \tag{A1}$$

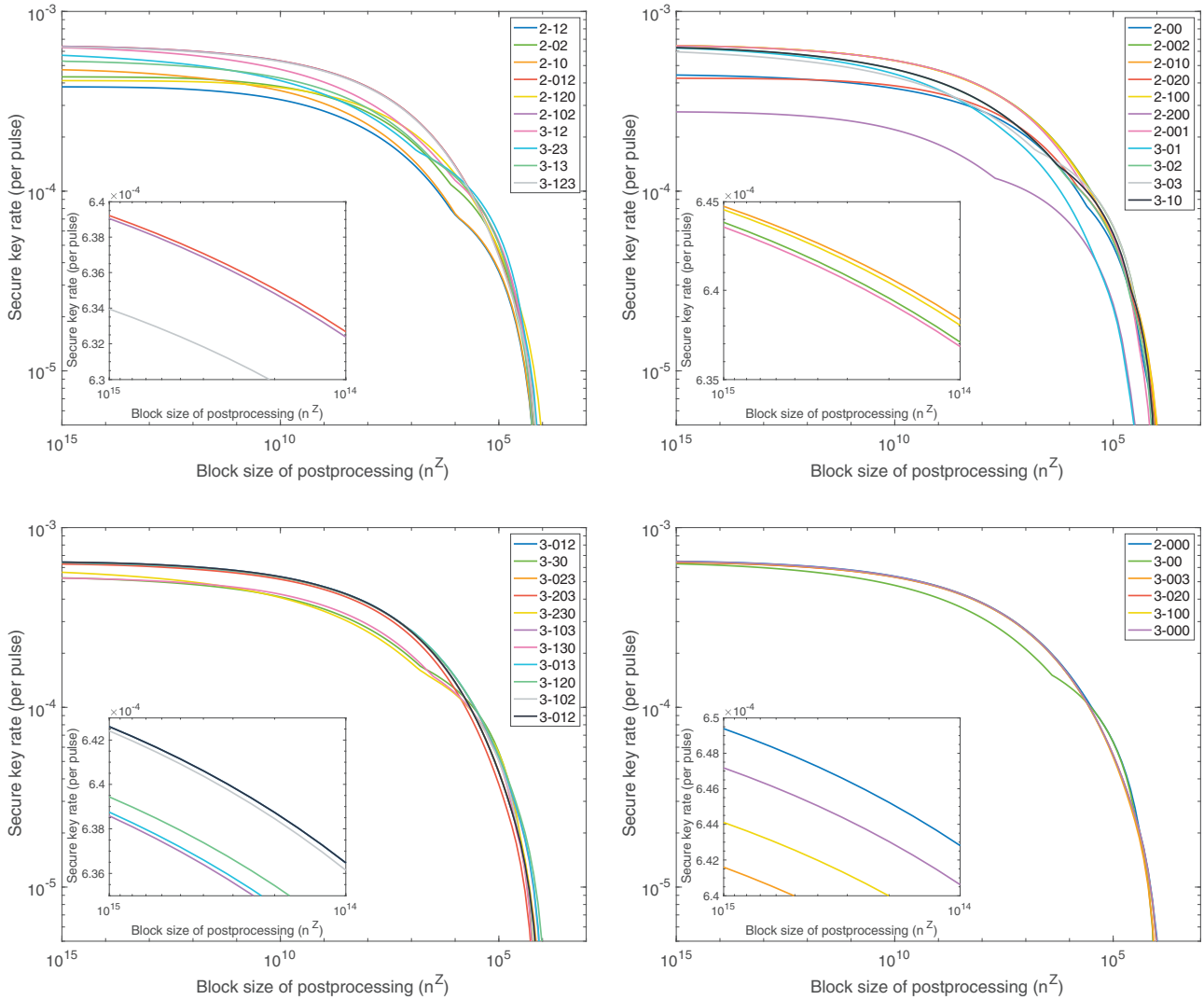


Figure B2. Secure key rates of the decoy-state protocols as functions of the block size of postpocessing, n^Z, listed in Table 1. Each figure has a subfigure to show the magnification of the local area at 0 dB. The digits of the legend located in the left and the right to the hyphen are the N_{IZ} and *Xlock*, respectively.

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where ω_a and ω_b are the basis choice of Alice and Bob, respectively, $Q^{\omega_b,}_{\mu_{\omega_a i}}$ and $T^{\omega_b}_{\mu_{\omega_a i}}$ are the overall yield and error yield.

Besides, the lower bound of single-photon events is given by

$$Y_1^{\omega_b,-} = \max\left\{\frac{p_{2|\mu_{\omega_a 2}} Q^{\omega_b}_{\mu_{\omega_a 1}} - p_{2|\mu_{\omega_a 1}} Q^{\omega_b}_{\mu_{\omega_a 2}} - P^{0,2}_{\mu_{\omega_a 1},\mu_{\omega_a 2}} Y_0^{\omega_b,+}}{P^{1,2}_{\mu_{\omega_a 1},\mu_{\omega_a 2}}} | \right.$$
$$\left. \omega_a \in \{X, Z\}\right\} \tag{A2}$$

where $P^{m,n}_{\mu_{\omega_a i},\mu_{\omega_a j}} = p_{m|\mu_{\omega_a i}} p_{n|\mu_{\omega_a j}} - p_{m|\mu_{\omega_a j}} p_{n|\mu_{\omega_a i}}$.

Also, the upper bounds of bit error rate is given by

$$e_1^{\omega_b,+} = \min\left\{\frac{p_{0|\mu_{\omega_a 1}} T^{\omega_b}_{\mu_{\omega_a 2}} - p_{0|\mu_{\omega_a 2}} T^{\omega_b}_{\mu_{\omega_a 1}}}{P^{0,1}_{\mu_{\omega_a 1},\mu_{\omega_a 2}} Y_1^{\omega_b,-}} | \omega_a \in \{X, Z\}\right\} \tag{A3}$$

The photon-number distributions of the coherent states with these two intensities satisfy

$$\frac{p_{k|\mu_{\omega_a 2}}}{p_{k|\mu_{\omega_a 1}}} \geq \frac{p_{1|\mu_{\omega_a 2}}}{p_{1|\mu_{\omega_a 1}}} \geq \frac{p_{0|\mu_{\omega_a 2}}}{p_{0|\mu_{\omega_a 1}}}. \tag{A4}$$

Besides, for the scenario that the number of decoy states preparing in ω_a basis is three, the lower bound of single-photon events is changed to

$$Y_1^{\omega_b,-} = \frac{\left[\frac{Q^{\omega_a,\omega_b}_{\mu_{\omega_a 2}}}{p_{0|\mu_{\omega_a 2}}} - \frac{Q^{\omega_a,\omega_b}_{\mu_{\omega_a 1}}}{p_{0|\mu_{\omega_a 1}}} - \left(\frac{Q^{\omega_a,\omega_b}_{\mu_{\omega_a 3}}}{p_{0|\mu_{\omega_a 3}}} - Y_0^{\omega_a,\omega_b,-}\right) \times \left(\frac{p_{2|\mu_{\omega_a 2}}}{p_{0|\mu_{\omega_a 2}}} - \frac{p_{2|\mu_{\omega_a 1}}}{p_{0|\mu_{\omega_a 1}}}\right)\frac{p_{0|\mu_{\omega_a 3}}}{p_{2|\mu_{\omega_a 3}}}\right]}{\left(\frac{p_{1|\mu_{\omega_a 2}}}{p_{0|\mu_{\omega_a 2}}} - \frac{p_{1|\mu_{\omega_a 1}}}{p_{0|\mu_{\omega_a 1}}}\right) - \left(\frac{p_{2|\mu_{\omega_a 2}}}{p_{0|\mu_{\omega_a 2}}} - \frac{p_{2|\mu_{\omega_a 1}}}{p_{0|\mu_{\omega_a 1}}}\right)\frac{p_{1|\mu_{\omega_a 3}}}{p_{2|\mu_{\omega_a 3}}}} \tag{A5}$$

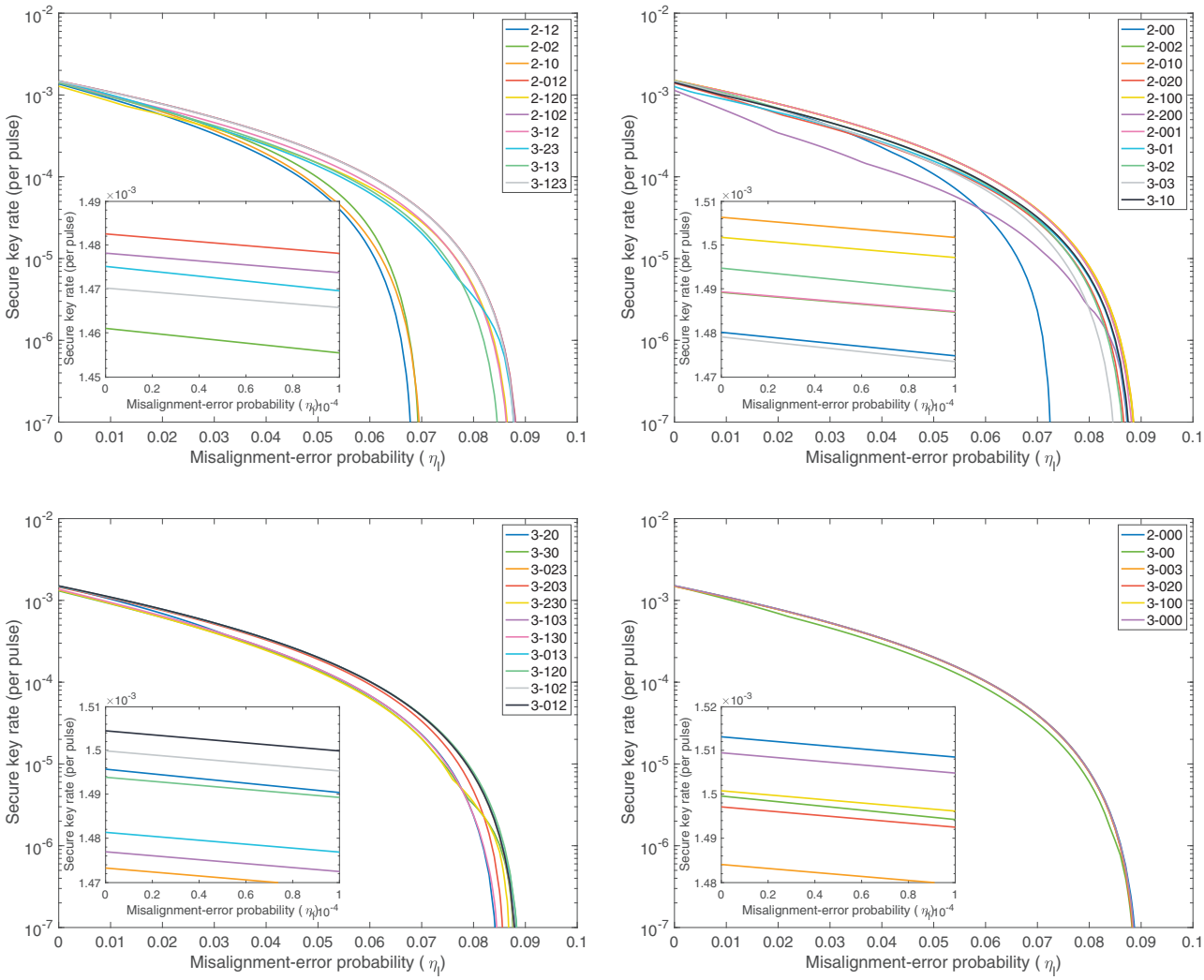


Figure B3. Secure key rates of the decoy-state protocols as functions of the probability of misalignment error, η_l, listed in Table 1. Each figure has a subfigure to show the magnification of the local area at 0 dB. The digits of the legend located in the left and the right to the hyphen are the N_{IZ} and *Xlock*, respectively.

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where the photon-number distributions of the coherent states also satisfy

$$\frac{\frac{p_{k|\mu_{\omega_a}2}}{p_{0|\mu_{\omega_a}2}}-\frac{p_{k|\mu_{\omega_a}1}}{p_{0|\mu_{\omega_a}1}}}{\frac{p_{k|\mu_{\omega_a}3}}{p_{0|\mu_{\omega_a}3}}} \leq \frac{\frac{p_{2|\mu_{\omega_a}2}}{p_{0|\mu_{\omega_a}2}}-\frac{p_{2|\mu_{\omega_a}1}}{p_{0|\mu_{\omega_a}1}}}{\frac{p_{2|\mu_{\omega_a}3}}{p_{0|\mu_{\omega_a}3}}} \leq \frac{\frac{p_{1|\mu_{\omega_a}2}}{p_{0|\mu_{\omega_a}2}}-\frac{p_{1|\mu_{\omega_a}1}}{p_{0|\mu_{\omega_a}1}}}{\frac{p_{1|\mu_{\omega_a}3}}{p_{0|\mu_{\omega_a}3}}} \tag{A6}$$

Finally, the numbers of the events can be obtained by

$$\begin{aligned} s_0^{\omega_b,-} &= N\tau_{0|\omega_a}p^{\omega_b}Y_0^{\omega_b,-} \\ s_1^{\omega_b,-} &= N\tau_{1|\omega_a}p^{\omega_b}Y_1^{\omega_b,-} \end{aligned} \tag{A7}$$

where N is the total number of the pulses sent by Alice, p_{ω_b} is the probability that Bob selects the ω_b basis, $\tau_{0|\omega_a}=\sum_n p_{\mu_{\omega_a n}}p_{0|\mu_{\omega_a n}}$, $\tau_{1|\omega_a}=\sum_n p_{\mu_{\omega_a n}}p_{1|\mu_{\omega_a n}}$.

Moreover,

$$\begin{aligned} n_{\mu_{\omega_a i}}^{\omega_b\pm} &= n_{\mu_{\omega_a i}}^{\omega_b} \pm \sqrt{\frac{n_{\omega_a}^{\omega_b}}{2}\ln\frac{1}{\tilde{\varepsilon}_{sec}}} \\ m_{\mu_{\omega_a i}}^{\omega_b\pm} &= m_{\mu_{\omega_a i}}^{\omega_b} \pm \sqrt{\frac{m_{\omega_a}^{\omega_b}}{2}\ln\frac{1}{\tilde{\varepsilon}_{sec}}} \end{aligned} \tag{A8}$$

where $\tilde{\varepsilon}_{sec}$ is the probability of failing to estimate, $n_{\mu_{\omega_a i}}^{\omega_b}$ and $m_{\mu_{\omega_a i}}^{\omega_b}$ are the counts of valid events and error events and can be derived by

$$\begin{aligned} n_{\mu_{\omega_a i}}^{\omega_b} &= Np_{\mu_{\omega_a i}}p_{\omega_b}Q_{\mu_{\omega_a}}^{\omega_b} \\ m_{\mu_{\omega_a i}}^{\omega_b} &= Np_{\mu_{\omega_a i}}p_{\omega_b}T_{\mu_{\omega_a}}^{\omega_b} \\ n_{\omega_a}^{\omega_b} &= \sum_i n_{\mu_{\omega_a i}}^{\omega_b} \\ m_{\omega_a}^{\omega_b} &= \sum_i m_{\mu_{\omega_a i}}^{\omega_b} \end{aligned} \tag{A9}$$

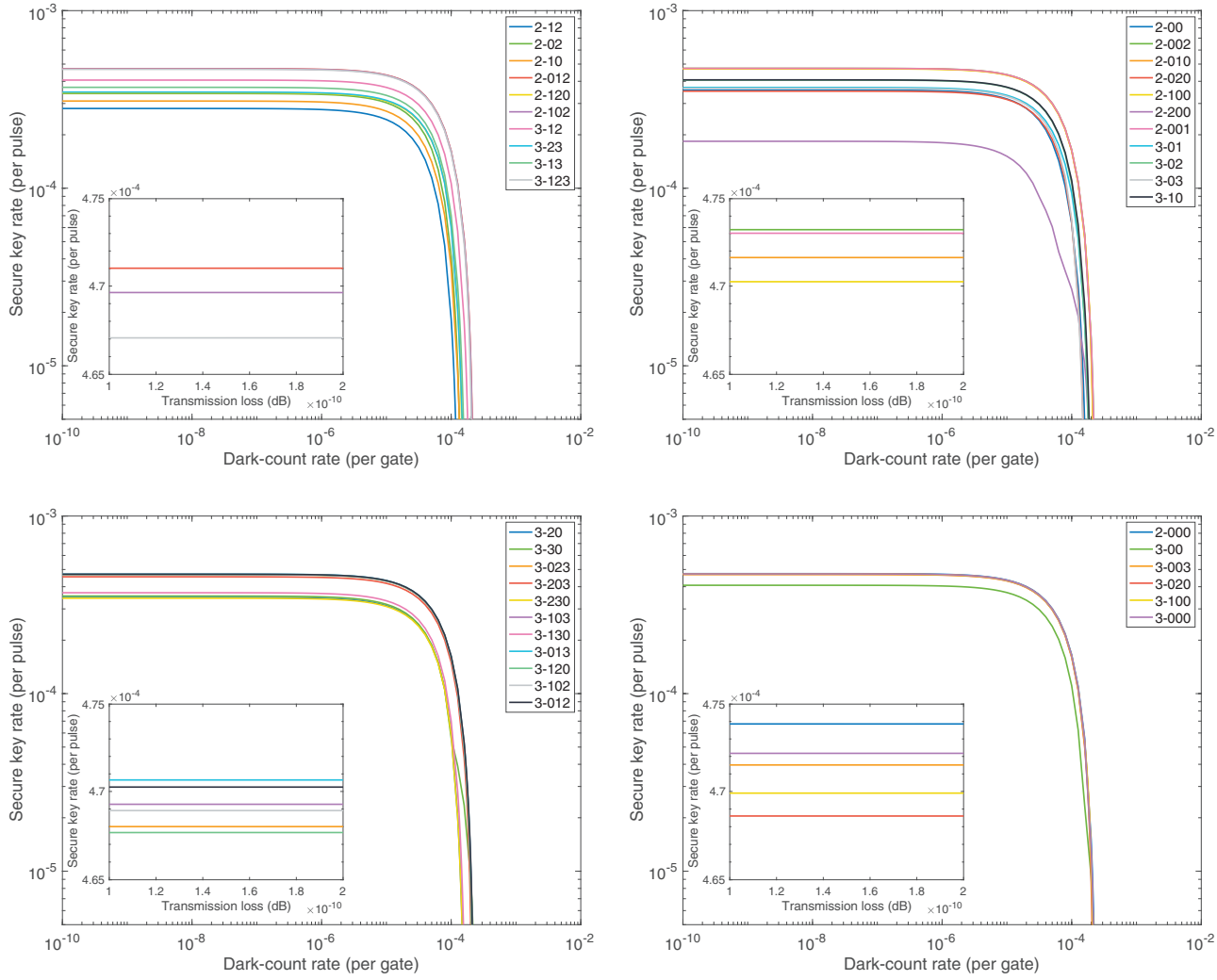


Figure B4. Secure key rates of the decoy-state protocols as functions of the dark-count probability, p_d, listed in Table 1. Each figure has a subfigure to show the magnification of the local area at 0 dB. The digits of the legend located in the left and the right to the hyphen are the N_{IZ} and *Xlock*, respectively.

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According to the analysis in ref. [39], the relationship between them satisfy

$$e_{1,p}^{\omega+} = e_1^{\bar{\omega}+} + g\left(\tilde{\varepsilon}_{sec}, e_1^{\bar{\omega}+}, s_1^{\bar{\omega}-}, s_1^{\omega-}\right)$$

$$g(a,b,c,d) = \sqrt{\frac{b(1-b)(c+d)\ln 2}{cd}\log_2\sqrt{\frac{c+d}{b(1-b)cd}\frac{1}{a^2}}} \quad \text{(A10)}$$

where ω and $\overline{\omega}$ are different bases, that is, $\omega = Z$ when $\overline{\omega} = X$ and vice versa.

Appendix B: Secure Key Rates of the Decoy-State Protocols

We first provide the secure key rates as functions of the transmission loss in **Figure B1**. Because there are too many schemes simulated, they are split into four groups by the columns of Table 2 for readability.

Influenced by the two-basis key generation, there are one or two inflexions in some key-rate lines. The first inflexion happens when the secure key rate in Z basis starts to be lower than it obtained in X basis. The second inflexion represents one of the bases cannot generate a positive key. Both cases can change the gradient of secure key rates.

Figure B2 shows the secure key rates of various decoy-state protocols shown in Table 1. The direct constriction of the block size magnifies the finite-size effect. Therefore, compared with Figure B1, there are more inflexions due to the more significant effect.

Finally, the effects of the misalignment-error probability, dark-count probability, and afterpulse rate on the secure key rates are shown in **Figures B3–B5**, respectively. At the transmission loss of 10 dB, the tolerances to η_l and p_d can be up to 9% and 10^{-4}, respectively.

As for the afterpulse effect, most decoy-state protocols can generate positive key over 50% of afterpulse rate, where the intensities of decoy states have decreased to a low value. Therefore, the secure key rates as functions of afterpulse rate do not present abrupt falls because the intensities have met the limits.

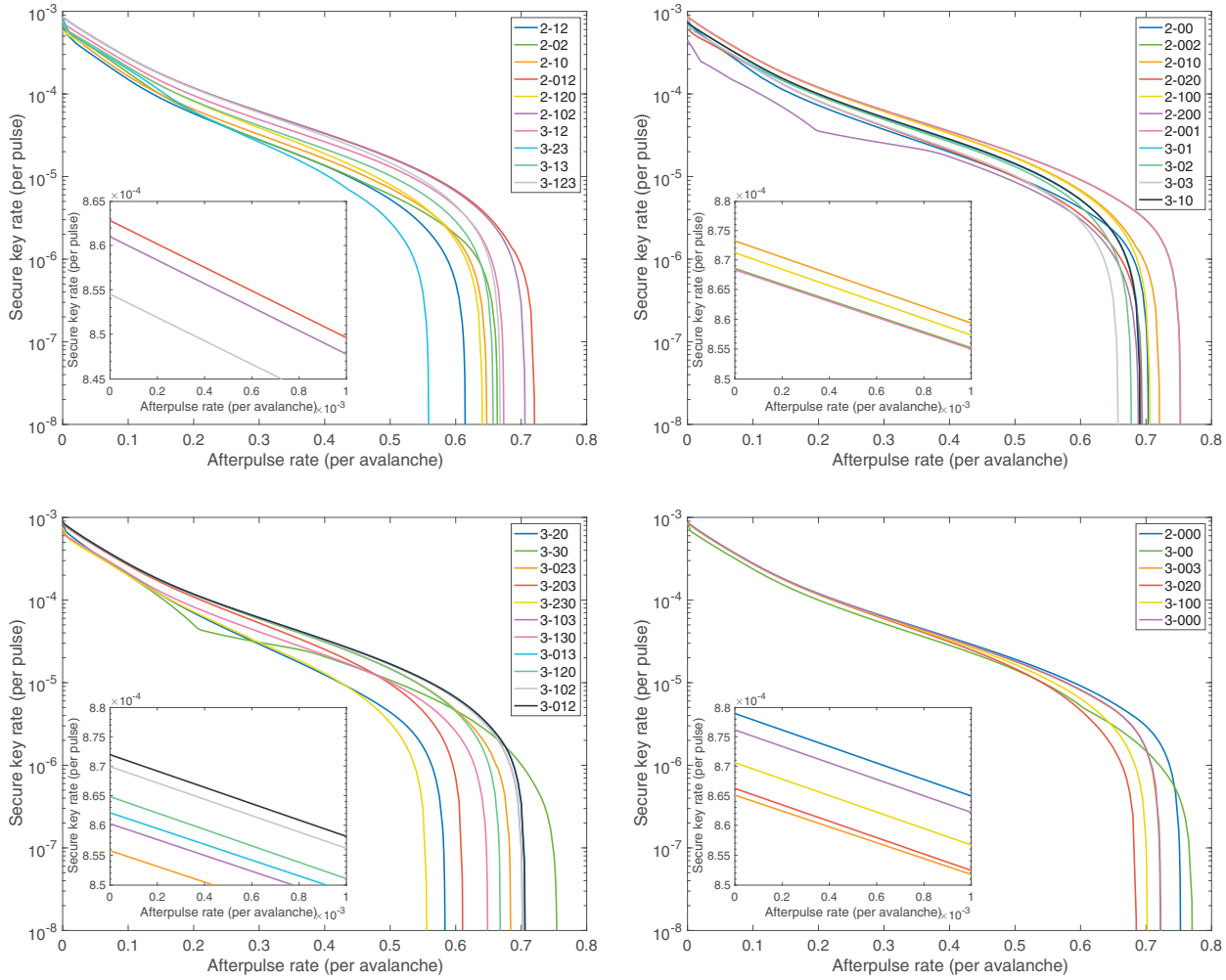


Figure B5. Secure key rates of the decoy-state protocols as functions of the afterpulse rate, p_a, listed in Table 1. Each figure has a subfigure to show the magnification of the local area at 0 dB. The digits of the legend located in the left and the right to the hyphen are the N_{IZ} and *Xlock*, respectively.

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Acknowledgements

The authors gratefully acknowledge financial support from the National Key Research and Development Program of China (Grant No. 2018YFA0306400), the National Natural Science Foundation of China (Grant Nos. 61622506, 61575183, 61627820, 61475148, and 61675189), and the Anhui Initiative in Quantum Information Technologies.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

decoy-state protocol, optimization algorithm, quantum key distribution, system modelling

Received: November 14, 2020
Revised: December 22, 2020
Published online: February 1, 2021

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